CODEX FOR EMPIRICAL RESEARCH

Work faster—
without outsourcing judgment

A two-hour course for data analysts and applied researchers

Method. Execution boundary. Audit.

Crash Course Data Analysis | 16 September 2026

The outcome is a working method

By the end, participants should be able to…

1. **Choose the right surface, model, and reasoning effort.**
2. **Structure tasks to conserve context and included usage.**
3. **Turn repeated research procedures into skills.**
4. **Keep execution and audit responsibilities explicit.**

Six sections, one research loop

A **Start**
Surfaces, install, access

B **Budget**
Tokens, limits, models

C **Manage**
Compaction and resets

D **Steer**
Voice and Remote

E **Reuse**
Skills and MCPs

F **Apply**
Three real projects

The course moves from product mechanics to a defensible research workflow.

A

Basic Codex functions

Where will Codex work, and how will access be paid for?

Choose the surface around the work

DESKTOP

Research artifacts

Files, PDFs, notebooks, slides, spreadsheets, voice, and local projects.

Best starting point for this course.

CLI / IDE

Code-first work

Repositories, scripts, tests, diffs, commands, and repeatable developer workflows.

Best when execution lives in a codebase.

CLOUD

Hosted execution

Background tasks and connected repositories without keeping the local session active.

Best when the environment is already reproducible.

Install the surface you will actually use

Product details verified 16 September 2026

DESKTOP APP

1 Download the ChatGPT desktop app for your operating system.
2 Sign in with ChatGPT or an available alternative.
3 Open a folder or create a project.

CODEX CLI

1 Use the official installer shown on the CLI page.
2 Run codex from the project directory.
3 Confirm model, permissions, and writable roots.

macOS / Linux installer
curl -fsSL https://chatgpt.com/codex/install.sh | sh

Install once; verify updates and commands against the official page before teaching.

Start with a bounded project

01

Select folder

Only the files needed for the task

02

State goal

One concrete deliverable

03

Set boundary

What Codex may change or run

04

Define check

How success will be verified

Example

“Create a separate Stata do-file. Do not overwrite the baseline. Perform static checks only; I will run Stata and return the log.”

Two access routes

CHATGPT SIGN-IN

Subscription access

Included allowance
Workspace policies and features
Shared limits across eligible agentic work

Good for

Interactive research and local projects

API KEY

Usage-based access

Input and output token billing
Programmatic control
Some ChatGPT workspace features unavailable

Good for

Automation, CI, and metered workflows

Pick the account around governance and workload

Plan names and allowances change—check the current pricing page before delivery

INDIVIDUAL	**Free / Go / Plus / Pro**	Personal work; included usage varies by plan.
TEAM	**Business**	Shared workspace, controls, and pooled or plan-based usage.
INSTITUTION	**Enterprise / Edu**	Administration, governance, and negotiated access.
METERED	**API key**	Usage-based billing for local or automated workflows.

B

Token usage and limits

How do we spend less without making the work worse?

Prompt length is only one usage driver

MODEL

Capability and rate

CONTEXT

History and attached files

REASONING

Depth of internal work

TOOLS

Search, files, browser, MCP

TASK

Complexity and iteration

SURFACE

Local or cloud execution

Two similar-looking prompts can consume very different amounts of included usage.

Three meters—do not confuse them

TOKENS

Information processed

Prompt, files, history, tool results, and response all use tokens.

A technical unit.

CREDITS

Eligible paid usage

A billing unit used on credit-based plans or after included limits are reached.

A commercial unit.

CONTEXT

What the model carries

The active working memory available for this conversation and its evidence.

A quality constraint.

Treat the dashboard as the source of truth

5-hour window

Short-horizon included usage

Weekly window

Longer-horizon allowance may also apply

USAGE CHECK

Desktop / web
Usage dashboard

CLI
/status and /usage

Do not plan a workshop around an old screenshot of plan limits.

Use the smallest model that meets the risk

Model roles verified 16 September 2026

GPT-6 ASTRA	**Hardest end-to-end work**	Ambiguous research design, complex audits, computer use
GPT-5.6 SOL	**Complex professional work**	Demanding coding, document production, deep analysis
GPT-5.6 TERRA	**Balance intelligence and cost**	Everyday analysis, reporting, focused coding
GPT-5.6 LUNA	**Cost-sensitive volume**	Extraction, classification, routine transforms

Reasoning effort is a separate dial

LOW

Formatting
Straightforward extraction
Small repairs

MEDIUM

Routine analysis
Document drafting
Standard debugging

HIGH+

Ambiguous specifications
Difficult causal logic
High-stakes audit

faster / cheaper → **more deliberate**

Escalate effort when uncertainty or consequence rises—not because the task is long.

A precise brief prevents expensive wandering

GOAL	What must exist when the task is done?
CONTEXT	Which files, evidence, and prior decisions matter?
OUTPUT	Format, audience, length, and location.
BOUNDARIES	What may change, run, publish, or remain untouched?
CHECK	What evidence proves the work is correct?

Seven habits that make limits last longer

1. **Point to relevant files—not the whole drive.**
2. **Separate required work from optional improvements.**
3. **Ask for a plan before a long multi-step build.**
4. **Use Terra or Luna for routine, high-volume work.**
5. **Disable MCP servers you do not need.**
6. **Move repeated instructions into a skill.**
7. **Return errors, logs, and evidence—not a paraphrase.**



Compacting and resets

When should we preserve context, summarize it, or start fresh?

Four different decisions

CONTINUE	Same objective, same evidence, same deliverable.
COMPACT	Same objective, but history is long or noisy.
NEW TASK	Objective, dataset, or deliverable has changed.
RESET	Restore an eligible usage window; context does not improve.

A usage reset does not rescue an incoherent thread.

Compaction keeps decisions—not every detail

BEFORE

Requirements
Exploration
Tool output
Error logs
Intermediate drafts
Decisions



AFTER /COMPACT

Goal
Constraints
Key evidence
Decisions
Open questions

Discarded detail

Verbose logs
Dead ends
Redundant drafts
Transient tool noise

Compact after a meaningful milestone—then verify the summary kept the right constraints.

A reset restores capacity—when a window is eligible

01

Inspect

Usage dashboard or /usage

02

Confirm

An earned reset exists

03

Redeem

Apply to an eligible window

04

Refresh

Re-read current limits

A reset can return: reset, already redeemed, nothing to reset, or no credit.

Use a reset to recover capacity—not as a substitute for model choice, compaction, or scope control.

D

Voice commands

How can speech accelerate work without weakening the brief?

Dictation and Voice solve different problems

DICTATION

Speech → editable prompt

Press Ctrl+Shift+D with the composer visible.
Review the transcription before sending.

Best for

A long, precise task brief

VOICE CONVERSATION

Live dialogue with GPT-Live

Start, check, and steer work conversationally.
Tasks continue to use their selected model.

Best for

Coordination while away from the keyboard

Speak in research clauses

01

Objective

What should exist?

02

Evidence

Which files or sources?

03

Boundary

What may Codex do?

04

Audit

How will we check it?

VOICE EXAMPLE

“Use the Jamaica do-file as the baseline. Create a separate age-profile version. Do not run Stata. Check syntax statically, then tell me exactly what I should run and return.”

Remote keeps the work on the connected host

PHONE

Start
Steer
Approve
Review



CONNECTED HOST

Files
Credentials
Skills and plugins
Local tools

Remote is a control channel—not a copy of the project on the phone.

Remote keeps the safety boundary

HOST	Must remain connected and able to run the task.
APPROVALS	Permissions and approval prompts still apply.
WINDOWS	Computer use takes over the active foreground desktop.
SECRETS	Credentials stay on the connected host.
NETWORK	Do not expose the App Server publicly.

E

Skills and MCPs

Which parts of research should become reusable—and which need live access?

A skill defines the method; an MCP supplies access

SKILL

Reusable protocol

Screening rules
Required checks
Output structure
Templates and scripts

Example: replication audit

MCP

Controlled connection

Searchable literature
Repositories and drives
Databases and services
Permitted actions

Example: bibliographic search

Invoke the workflow explicitly when consistency matters

$presentation

Read the deck source, build editable slides, render every page, and repair layout defects.

$custom-replication-audit

Verify data lineage, specification, sample construction, outputs, and unresolved uncertainty.

Use a skill when you would otherwise repeat the same instructions, checks, and templates across tasks.

Literature review: automate the pipeline, audit the claims

01
Question
PICO / theory / scope

02
Search
Databases and web

03
Screen
Inclusion and exclusion

04
Extract
Design, sample, estimates

05
Synthesize
Map evidence and gaps

Skill **Protocol, screening fields, extraction schema, citation rules**

MCP **Authorized access to papers, drives, reference managers, or databases**

The researcher still checks relevance, design quality, and whether a citation supports the claim.

One audit logic across Stata, R, and Python

STATA

Do-files

Preserve baseline code. Create a separate variant. Check macros, paths, merges, samples, estimates, and exports.

Researcher runs Stata when required.

R

Scripts / Quarto

Pin packages and seeds. Separate raw, intermediate, and output files. Test joins, missingness, and model objects.

Render the document, not just the code.

PYTHON

Scripts / notebooks

Control environments. Assert schemas and types. Cache raw inputs. Log requests and test transformations.

Restart and run all cells.

A useful research skill contains the audit

TRIGGER	When should this workflow run?
INPUTS	Required files, metadata, and execution environment.
PROCEDURE	Ordered method and explicit boundaries.
CHECKS	Syntax, data, sample, model, output, and interpretation.
HANDOFF	What the researcher must run, inspect, or decide.

Connect only the systems the task needs

DISCOVER

What tools and data does the server expose?

AUDIT

What did Codex read, change, or send?

AUTHORIZE

Which account, scopes, and side effects are allowed?

DISABLE

Turn off unused servers to reduce risk and context cost.

More connections are not automatically more efficient.

F

Application examples

What changes when the work is code, a presentation, or a live data source?

Case 01 | Jamaica

Codex drafts; the researcher runs Stata

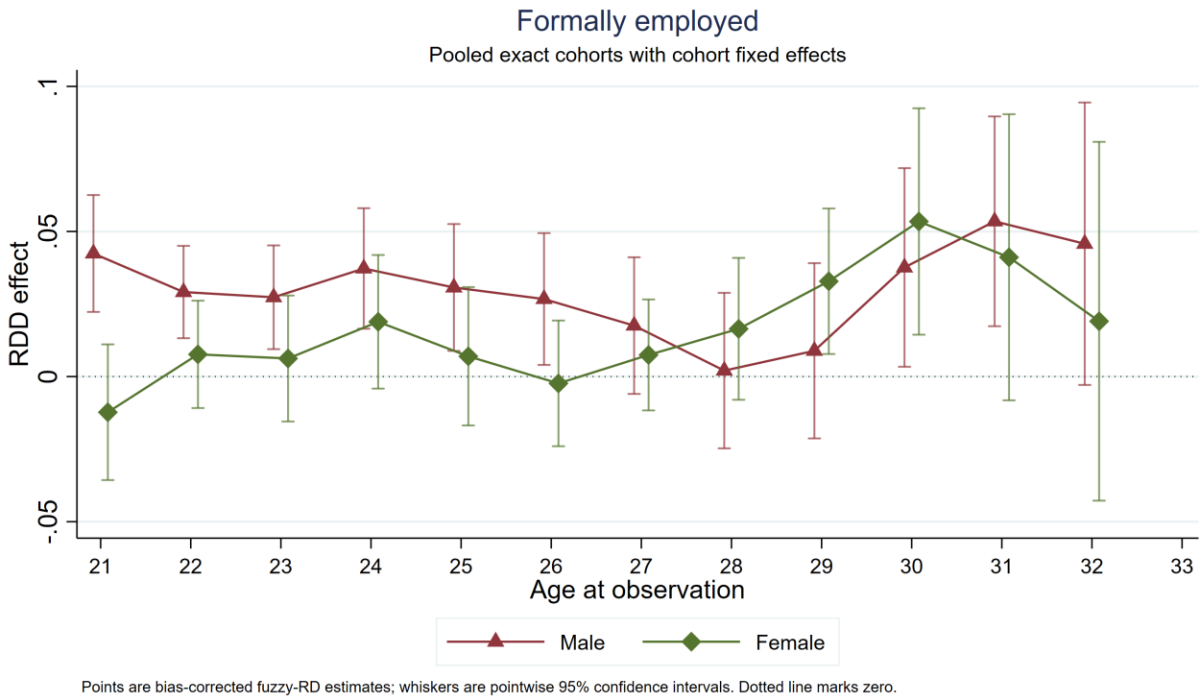
Speed comes from a clean execution boundary and a fast error-feedback loop.

The handoff is part of the method

1. **Codex creates a separate do-file and checks it statically.**

2. **The researcher runs Stata and returns the exact log or error.**

3. **Both audit specifications, realized samples, tables, and graphs.**



The execution boundary protects the empirical workflow while preserving speed.

The first Stata run exposed a hard limit

RUNTIME ERROR

```
. local outcome_variable_pctcontributions
"pct_contr"
_outcome_variable_pctcontributions
invalid name
```

DIAGNOSIS

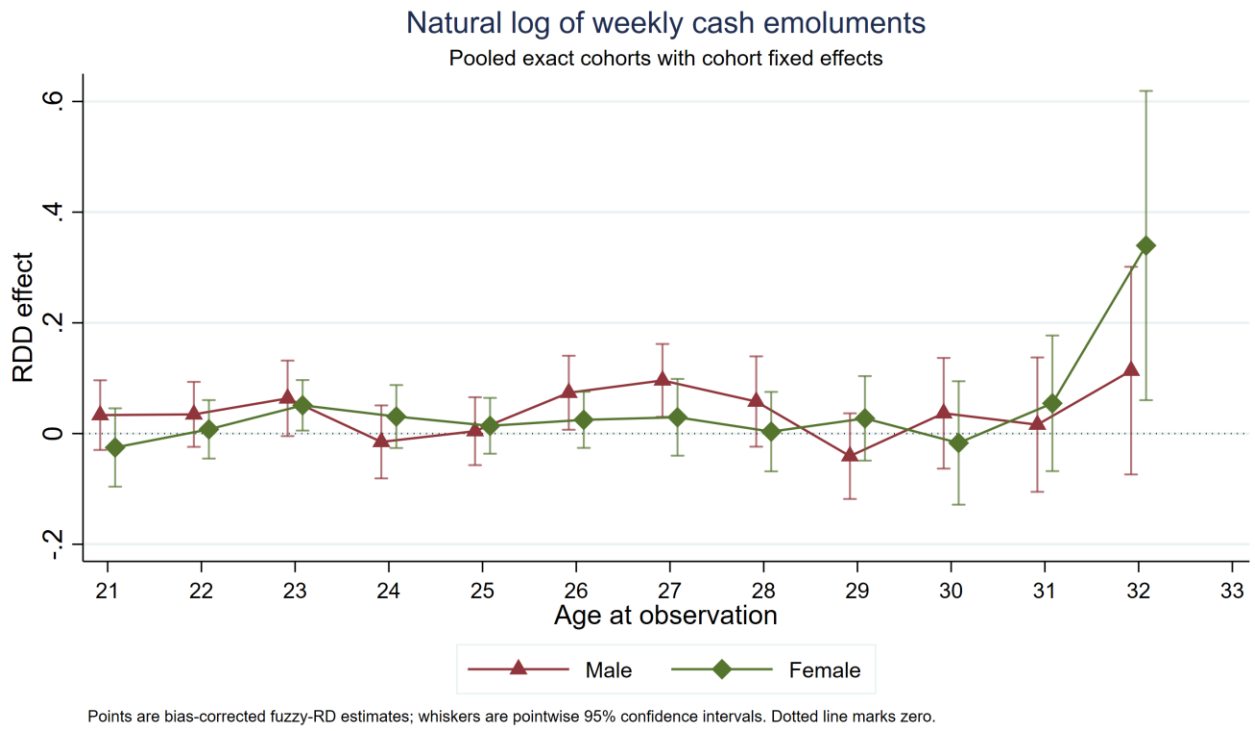
Stata macro names cannot exceed 31 characters.

REPAIR

```
local outcome_stem_pctcontributions
"pct_contr"
```

Static review reduced risk; only the real runtime revealed the environment-specific failure.

A successful run is not the end of the audit



THREE QUESTIONS

1 Did the code estimate the intended specification?

2 Which people, cohorts, and outcome years entered?

3 Does the interpretation match the design and uncertainty?

Same N can still mean a different age-year contribution.

Case 02 | Trinidad and Tobago

A report becomes a Ministry presentation

Codex can accelerate narrative and production—if every rendered slide is inspected.

Presentation work is a research translation task

SOURCE

Final report, figures, and policy audience

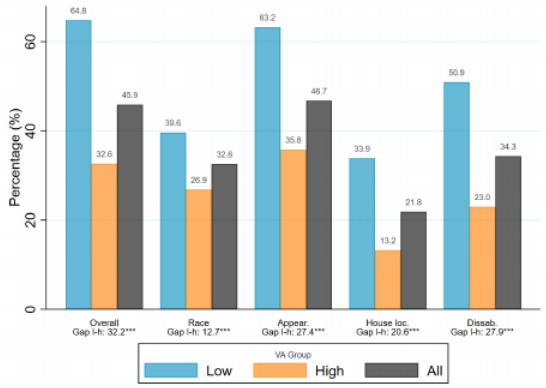
CRAFT

Narrative structure, Beamer source, and section transitions

AUDIT

Compile, render, inspect legends, revise

Safety is the clearest CSEC-VA divide



Bullying is a problem

64.8%	32.6%
Low VA	High VA

Students feel safe

21.1%	50.7%
Low VA	High VA

Less bullying and greater safety move together in high-VA schools.

Source: Final report, Figure 10 and Table 3. Low and high are CSEC VA terciles for 2012-2020.

Visual QA changed the final deck



Compiling is a technical check. Rendering every slide is a communication check.

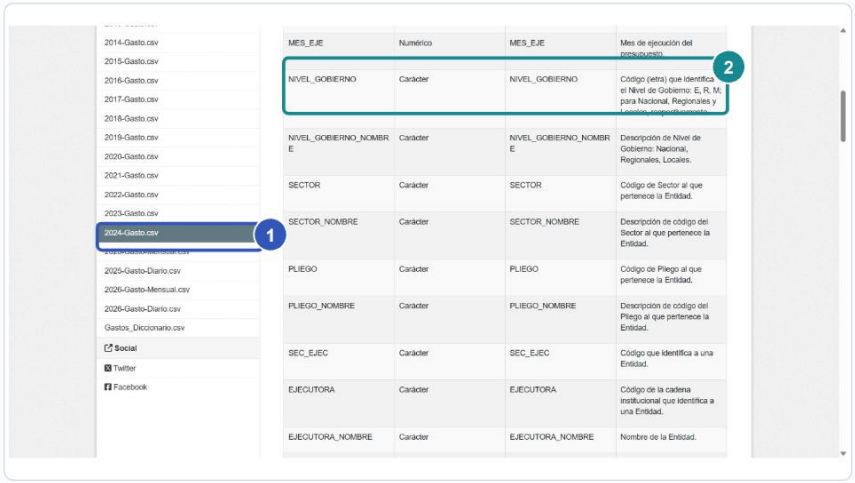
Case 03 | Peru web scraping

Notebooks first; slides second

The teaching material becomes credible only after the data-access paths are tested.

The notebooks build capability in layers

MEF resources and the API request



Notebook request

```
MEF_API = ".../datastore_search"

params = {
  "resource_id": RESOURCE_2024,
  "limit": 500,
  "offset": 0,
}
```

1 selects the annual data resource. 2 explains its columns.

WEBSITE CONFIGURATION

MEF provides downloadable annual files and an API. The notebook uses the API pattern for small batches and the bulk files for very large data.

8

SEQUENCE

00 HTTP, HTML, and APIs

01 MEF Consulta Amigable

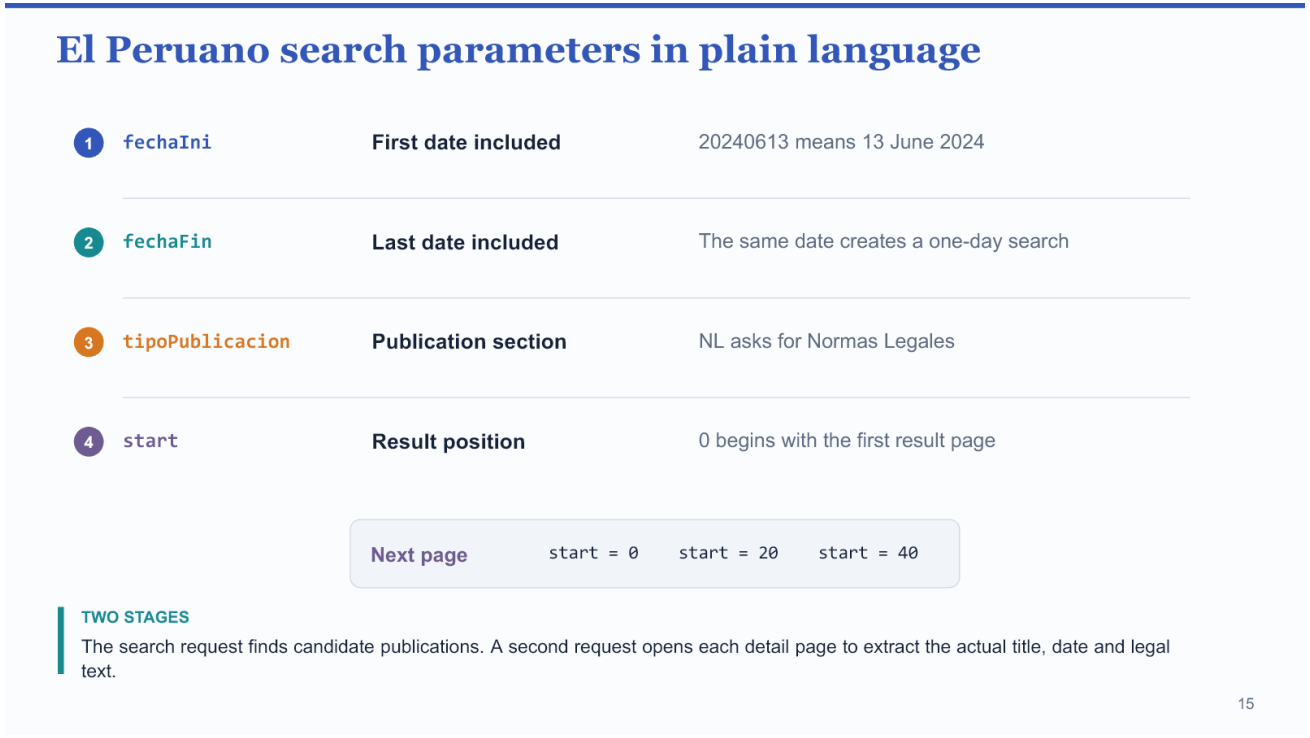
02 El Peruano norms

03 SEACE OCDS JSON

04 Integrate by CUI

Each notebook leaves a testable artifact for the next one.

Evaluate access before teaching the output



HTTP — Status code and content type

SCHEMA — Expected fields and types

ROWS — Counts and key uniqueness

CACHE — Raw response preserved

LINEAGE — Source manifest updated

Live-refresh switches keep the lesson reproducible when public endpoints change.

THE PRODUCTIVITY EQUATION

Speed × judgment × audit

Codex is most valuable when the researcher can specify the method, control the execution boundary, and recognize a plausible-but-wrong result.

1 **Bound the task**

2 **Match the model**

3 **Reuse the method**

4 **Inspect the evidence**

5 **Keep responsibility explicit**

What recurring workflow will you turn into a skill?